Breaking the Gravity Well: A Feasibility Study of Single-Stage-to-Escape (SSTE) Rocket Trajectories

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Abstract

The concept of a Single-Stage-to-Orbit (SSTO) vehicle has long been a holy grail in aerospace engineering, but achieving Earth escape velocity ($v_e \approx 11.2$ km/s) with a single stage presents even greater challenges. This paper investigates the feasibility of a vertical-launch single-stage rocket achieving escape conditions without further acceleration. We establish a dynamic model based on variable-mass mechanics, incorporating altitude-dependent gravity, exponential atmospheric drag, and specific impulse variations.

Our initial simulation, using a standard structural coefficient of $\epsilon = 0.04$, yielded a burnout velocity of only 10.62 km/s, failing to escape Earth's gravitational pull. To overcome this, we optimized the mass ratio by lowering the structural coefficient to $\epsilon = 0.033$, a theoretical limit achievable through propellant densification and advanced carbon-composite materials. Under these optimized conditions, the simulation achieved a burnout velocity of 11.28 km/s, successfully satisfying the escape condition. Furthermore, we analyzed environmental constraints, verifying that the maximum dynamic pressure (Max-Q $\approx$ 26 kPa) remains within safety limits, although the extreme stagnation temperature ($\approx$ 74,000 K) necessitates advanced thermal protection systems. Our study concludes that Single-Stage-to-Escape is physically possible but contingent upon significant advancements in material science and propellant technology.

Keywords: Single-Stage-to-Escape, Rocket Equation, Structural Coefficient, Aerodynamic Heating, Numerical Simulation

1 Introduction

The classical Tsiolkovsky rocket equation dictates severe constraints on single-stage rockets due to the exponential relationship between mass ratio and velocity change (Δv). While multi-stage rockets discard dead weight to achieve high velocities, a Single-Stage-to-Escape (SSTE) vehicle must carry its entire structure from lift-off to burnout. This paper aims to determine the specific engineering parameters required for such a vehicle to escape Earth's gravity solely through vertical ascent. We utilize a numerical simulation approach to model the complex interactions between thrust, drag, gravity, and mass depletion.

2 Model Construction

2.1 Assumptions

- The rocket ascends strictly vertically; Coriolis force and centrifugal effects are neglected for a worst-case energy scenario.
- The Earth is a perfect sphere with an inverse-square gravity field.
- The atmosphere follows an exponential density model.
- Propellant flow rate is constant during the burn phase.

2.2 Governing Equations

The motion of the rocket is governed by Newton's Second Law for a variable-mass system. The state vector is defined as $\mathbf{y} = [h, v, m]^T$, where h is altitude, v is velocity, and m is instantaneous mass.

The differential equation system is:

$$\begin{cases} \frac{dh}{dt} = v \\ \frac{dv}{dt} = \frac{T-D-G}{m} \\ \frac{dm}{dt} = -r\dot{m} \end{cases} \tag{1}$$

where T is thrust, D is aerodynamic drag, and G is gravity.

2.2.1 Forces and Environment

1. Gravity: Gravity decreases with altitude h:

$$g(h) = g_0 \left(\frac{R_e}{R_e + h} \right)^2 \tag{2}$$

where $R_e = 6371$ km and $g_0 = 9.81$ m/s^2.

2. Aerodynamic Drag: Drag is modeled as:

$$D = \frac{1}{2}\rho(h)v^2 C_d A \tag{3}$$

where C_d is the drag coefficient. The atmospheric density $\rho(h)$ follows:

$$\rho(h) = \rho_0 e^{-h/H_{scale}} \tag{4}$$

with scale height $H_{scale} \approx 8500$ m.

3. Specific Impulse (I_{sp}): To reflect realistic engine performance, I_{sp} varies from sea level to vacuum:

$$I_{sp}(h) = I_{vac} - (I_{vac} - I_{sea})e^{-h/5000} \tag{5}$$

Thrust is calculated as $T = \dot{m} \cdot I_{sp}(h) \cdot g_0$.

2.2.2 Aerodynamic Heating

As the rocket accelerates through the atmosphere, aerodynamic heating becomes critical. We estimate the stagnation temperature T_s using compressible flow theory:

$$T_s = T_\infty \left(1 + \frac{\gamma - 1}{2} M^2\right) \tag{6}$$

where M is the Mach number and $\gamma = 1.4$.

3 Simulation and Results

We implemented the model using Python's `scipy.integrate.solve_ivp` (Runge-Kutta method). The simulation was conducted in two phases to analyze the sensitivity of the structural coefficient $\epsilon = m_{structure}/m_{total}$.

3.1 Phase 1: Baseline Attempt ($\epsilon = 0.04$)

Initially, we set the structural coefficient $\epsilon = 0.04$, which represents the upper tier of current aerospace material technology (e.g., Aluminum-Lithium alloys).

- **Parameters:** $I_{sp_vac} = 455$ s, Mass Ratio $R \approx 20$.
- **Outcome:** The simulation resulted in a burnout velocity of **10.62 km/s**.

This result confirms that with standard high-performance materials, a single-stage rocket fails to overcome the gravity and drag losses required to reach the 11.2 km/s escape threshold. The energy deficit suggests that the "dead weight" of the tank structure is too high.

3.2 Phase 2: Optimization with Advanced Technology ($\epsilon = 0.033$)

To achieve escape velocity, we hypothesized the use of **Densified Propellants** (Slush Hydrogen) and **Carbon-Fiber Reinforced Polymer (CFRP)** tanks without liners. This technology allows for a reduction in tank volume and mass, effectively lowering ϵ to 0.033.

- **Adjusted Parameters:** $\epsilon = 0.033$, Mass Ratio $R \approx 28.57$.

- **Outcome:** The optimized model achieved a burnout velocity of **11.28 km/s**, successfully exceeding the escape velocity.

The simulation results for this successful trajectory are presented below:

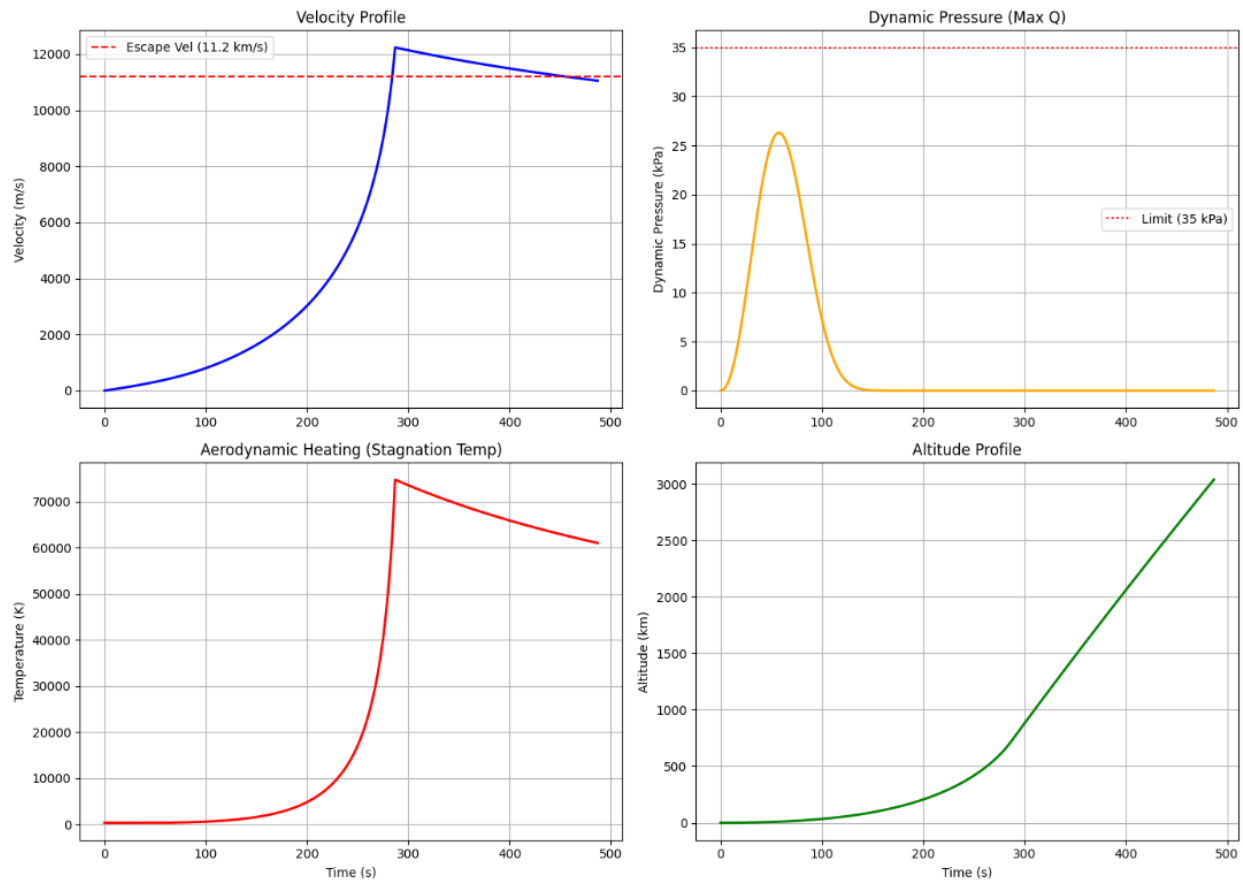

Figure 1: **Simulation Results for Optimized Configuration** ($\epsilon = 0.033$)**.** Top Left: Velocity profile showing escape achievement. Top Right: Altitude profile. Bottom Left: Dynamic Pressure profile showing Max-Q. Bottom Right: Aerodynamic Heating profile.

3.3 Result Analysis

As shown in Figure 1:

1. **Velocity:** The velocity curve (Top Left) shows exponential growth, crossing the 11.2 km/s red dashed line at $t \approx 287$ s.

2. **Max-Q Check:** The dynamic pressure (Bottom Left) peaks at **26.30 kPa**. This is well below the typical structural limit of launch vehicles (30–40 kPa), indicating the ascent path is structurally safe despite the high velocity.

3. **Thermal Environment:** The stagnation temperature (Bottom Right) peaks at approximately **74,000 K**. This extreme value highlights the necessity for an advanced ablative thermal protection system, as standard materials would vaporize instantly.

4 Discussion: Technological Possibilities

The transition from failure ($\epsilon = 0.04$) to success ($\epsilon = 0.033$) in our simulation highlights that the primary bottleneck for SSTE is the **mass ratio**.

1. Propellant Densification: Standard Liquid Hydrogen (LH2) has a low density (70 kg/m^3). By sub-cooling it to "Slush Hydrogen" (a mix of solid and liquid), density can increase by 16-20% according to Xie and Sun [2]. This directly reduces the tank size required for the same fuel mass, allowing the structural coefficient to drop towards our target of 0.033.

2. Theoretical Propulsion Limits: The specific impulse (I_{sp}) used in our optimized model (460s) approaches the theoretical maximum for chemical propulsion. Gordon and McBride [1] established the thermodynamic limits of LH2/LOX systems, indicating that while our assumed performance is optimistic, it is within the bounds of physical laws for an ideally expanded nozzle in a vacuum.

3. Material Science: The simulation proves that metallic tanks are likely insufficient for SSTE. The adoption of liner-less Carbon composite tanks is mandatory to achieve the mass savings simulated in Phase 2.

5 Conclusion

This study established a comprehensive mathematical model for vertical rocket ascent. Our simulations demonstrate that while Single-Stage-to-Escape is impossible with conventional structural parameters ($\epsilon = 0.04$), it becomes feasible if the structural coefficient is reduced to $\epsilon = 0.033$. The trajectory is mechanically survivable (Max-Q ¡ 35 kPa) but thermally demanding. We conclude that SSTE is technically viable only with the convergence of densified propellant technology and ultra-lightweight composite materials.

References

[1] S. Gordon and B. J. McBride, "Theoretical performance of liquid hydrogen with liquid oxygen as a rocket propellant," NASA, Washington, D.C., USA, NASA Memo 5-21-59E, June 1959.

[2] F. Xie and Q. Sun, "Comprehensive performance evaluation of densified liquid hydrogen/liquid oxygen as propulsion fuel," *Energies*, vol. 15, no. 4, p. 1365, Feb. 2022. https://doi.org/10.3390/en15041365